



Storm Sequence Generator — Test Results

Automated Test Documentation for Model Governance

MKM Research Labs

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1 Test Results — Storm Sequence Generator (MKM-SS-001)

Tests run on **21 March 2026 at 13:30:41**.

Table 1: Test Results Summary — Storm Sequence Generator (MKM-SS-001)

Total Tests	52
Passed	52
Failed	0
Skipped	0
Duration	0.22s

Table 2: Detailed Test Results — Storm Sequence Generator

Test Description	Acceptance Criteria	Result	Time
Duration must not exceed MAX_PRECIP_END_HOUR (156).	<code>result.storms[0].duration_hours <= MAX_PRECIP_END_HOUR</code>	Pass	0.000s
Isolated has one storm	<code>isinstance(result, StormSequence);</code> <code>result.num_storms == 1 (+2 more)</code>	Pass	0.000s
All storm sequences	<code>isinstance(seq, StormSequence)</code>	Pass	0.000s
Catchment propagated	<code>seq.catchment_id == "rhine";</code> <code>storm.catchment_id == "rhine"</code>	Pass	0.001s
Correct count	<code>len(small_batch) == 200</code>	Pass	0.000s
Different seeds differ	<code>types_a != types_b</code>	Pass	0.002s
Force sequence type doublet	<code>doublets > 50; seq.sequence_type in ("doublet", "isolated")</code>	Pass	0.002s
Force sequence type isolated	<code>seq.sequence_type == "isolated";</code> <code>seq.num_storms == 1</code>	Pass	0.001s
Reproducible with same seed	<code>[s.sequence_id for s in a] != [s.sequence_id for s in b];</code> <code>sa.num_storms == sb.num_storms</code> (+1 more)	Pass	0.003s
Returns list	<code>isinstance(small_batch, list)</code>	Pass	0.000s
Custom weight heavy on extreme shifts the distribution.	<code>all(c == "extreme" for c in cats)</code>	Pass	0.007s
All four default weight categories appear in the batch.	<code>cat in categories</code>	Pass	0.000s
Multi-storm sequences appear at a meaningful rate for severe+.	<code>multi_frac > 0.15</code>	Pass	0.000s
Creates parent dirs	<code>out.exists()</code>	Pass	0.008s
Load missing file raises	—	Pass	0.000s
Load returns correct count	<code>len(loaded) == len(small_batch)</code>	Pass	0.020s

Test Description	Acceptance Criteria	Result	Time
Load wrong schema raises	—	Pass	0.001s
Loaded sequences still valid	<code>result["invalid"] == 0</code>	Pass	0.021s
Round trip num storms	<code>orig.num_storms == back.num_storms</code>	Pass	0.022s
Round trip precipitation	<code>abs(orig.total_precipitation_mm - back.total_precipitatio...</code>	Pass	0.023s
Round trip sequence type	<code>orig.sequence_type == back.sequence_type</code>	Pass	0.021s
Round trip storm fields	<code>os_.storm_index == bs_.storm_index; abs(os_.duration_hours - bs_.duration_hours) < 0.01</code> (+1 more)	Pass	0.023s
Round trip total duration	<code>abs(orig.total_duration_hours - back.total_duration_hours...</code>	Pass	0.021s
Save creates file	<code>out.exists()</code>	Pass	0.008s
Schema version in file	<code>raw["schema_version"] == SCHEMA_VERSION</code>	Pass	0.020s
Creates file	<code>out.exists()</code>	Pass	0.000s
Duration stats present	<code>d["min"] > 0; d["max"] <= 156</code> (+1 more)	Pass	0.001s
Num sequences matches	<code>data["num_sequences"] == len(small.batch)</code>	Pass	0.001s
Precipitation stats present	<code>p["min"] > 0; p["max"] >= p["min"]</code> (+1 more)	Pass	0.001s
Schema version	<code>data["schema_version"] == SCHEMA_VERSION</code>	Pass	0.002s
Type counts sum to total	<code>sum(data["sequence_type_counts"].values()) == len(small.b...</code>	Pass	0.002s
Type fractions sum to one	<code>abs(total_frac - 1.0) < 0.001</code>	Pass	0.001s
Aggregate stats	<code>result.max_intensity_factor >= result.avg_intensity_factor; result.total_precipitation_mm > 0</code> (+1 more)	Pass	0.000s
Catchment id in storms	<code>result.storms[0].catchment_id == "rhine"</code>	Pass	0.000s
Cluster sequence	<code>result.sequence_type in (SequenceType.CLUSTER.value, Sequ...</code>	Pass	0.000s
force_sequence_type='doublet' → always doublet.	<code>result.sequence_type == SequenceType.DOUBLET.value; result.num_storms == 2</code>	Pass	0.000s

Test Description	Acceptance Criteria	Result	Time
force_sequence_type='isolated' → always isolated.	<code>result.sequence_type ==</code> <code>SequenceType.ISOLATED.value;</code> <code>result.num_storms == 1</code>	Pass	0.000s
Minimal intensity (5% sequence probability) mostly produces isolated.	<code>isolated >= 10</code>	Pass	0.001s
Generate multiple sequences without error.	<code>isinstance(result, StormSequence);</code> <code>result.num_storms >= 1</code>	Pass	0.000s
Persistent sequence	<code>result.sequence_type in</code> <code>(SequenceType.PERSISTENT.value,</code> <code>S...</code>	Pass	0.000s
Returns storm sequence	<code>isinstance(result, StormSequence)</code>	Pass	0.000s
All sequences must have total_duration ≤ 168h.	<code>result.total_duration_hours <= 168.0</code>	Pass	0.000s
Storms have correct fields	<code>storm.duration_hours > 0;</code> <code>storm.intensity_factor > 0</code> (+2 more)	Pass	0.000s
Custom catchment	<code>gen.catchment_id == "rhine";</code> <code>gen.base_precipitation ==</code> <code>CATCHMENT_BASE_PRECIP["rhine"]</code>	Pass	0.000s
Default catchment	<code>gen.catchment_id == "thames";</code> <code>gen.base_precipitation ==</code> <code>CATCHMENT_BASE_PRECIP["thames"]</code>	Pass	0.000s
Same seed → same sequence.	<code>seq1.num_storms == seq2.num_storms;</code> <code>seq1.sequence_type ==</code> <code>seq2.sequence_type</code>	Pass	0.000s
Unknown catchment defaults to 35	<code>gen.base_precipitation == 35.0</code>	Pass	0.000s
Zero validation errors across the whole batch.	<code>result["invalid"] == 0</code>	Pass	0.000s
Medium batch all valid	<code>result["invalid"] == 0</code>	Pass	0.001s
No negative durations	<code>storm.duration_hours > 0</code>	Pass	0.000s
No negative precipitation	<code>seq.total_precipitation_mm > 0;</code> <code>storm.precipitation_mm > 0</code>	Pass	0.000s
Last storm precipitation must end by hour 156 (1e-9 FP tolerance).	<code>last_end <= MAX_PRECIP_END_HOUR +</code> <code>_FP_TOL</code>	Pass	0.000s

1.1 Test Document History

Date	Version	Description	Author
12-Mar-2026	1.0	Initial test suite (116 tests)	D. Kelly
21-Mar-2026	1.1	64 test(s) removed (total: 52)	D. Kelly